MULTI-CLOUD FDE STACK #08

MULTI-CLOUD FDE STACK · ISSUE 08 OF 08

Ship to any cloud from one pipeline.

Docker, GitHub Actions, Terraform and Kubernetes, without stored keys.

08

AR

Akhilesh Ranjan Singh

Senior DevSecOps Engineer · github.com/iarsingh

Swipe →

Build once, promote

1. PR: lint, tests, build
2. Merge: build + scan image
3. Tag with commit SHA
4. Deploy dev, run evals
5. Approve staging, then prod

AR
Akhilesh Ranjan Singh
github.com/iarsingh

Keyless auth from Actions

CLOUD	ACTION	TRUST
AWS	configure-aws-credentials	IAM role, OIDC
GCP	google-github-actions/auth	Workload Identity Fed.
Azure	azure/login	Federated credential

AR
Akhilesh Ranjan Singh
github.com/iarsingh

Terraform per cloud

01 Shared interface

Same inputs and outputs for every cloud module.

02 Remote state

S3, GCS or Azure Storage, with locking.

03 Plan in CI

Apply only from the pipeline, never a laptop.

04 Drift check

A scheduled plan that alerts on changes.

AR
Akhilesh Ranjan Singh
github.com/iarsingh

Pipeline hardening

DO

- ✓ permissions: read by default
- ✓ Pin actions to a commit SHA
- ✓ Environments with reviewers
- ✓ Concurrency per environment

AVOID

- ✗ Cloud keys in repo secrets
- ✗ Untrusted input in run scripts
- ✗ Rebuild the image per stage
- ✗ terraform apply from a laptop

AR
Akhilesh Ranjan Singh
github.com/iarsingh

Delivery review

- [x] OIDC to every cloud
- [x] Same image in every stage
- [x] Evals gate the promotion
- [x] Rollback is one command
- [x] Plan output reviewed on PRs

AR
Akhilesh Ranjan Singh
github.com/iarsingh

KEY TAKEAWAY

Promote the same image. Change only the cloud edge.

→ Federate GitHub Actions to each cloud

→ One Terraform interface, many modules

→ Never apply from a laptop

YOUR TURN

Does your pipeline still use a long-lived cloud key?

FOLLOW AKHILESH RANJAN SINGH

Series complete — save it for your next review

Save · Repost

AR
Akhilesh Ranjan Singh
github.com/iarsingh